

## Practical 7

\* Aim:- Write a PL/SQL program using FOR loop to insert ten rows into a database table.

\* Theory:- PL/SQL (Procedural language / Structured Query language) is Oracle's procedural extension of SQL that combines the data manipulation capabilities of SQL with programming features such as loops, conditions, variables and exception handling.

A For loop is an iterative control structure that repeatedly executes a set of statements for a specified number of times. The loop variable is automatically initialized and incremented by Oracle.

In large database applications, inserting multiple records manually can be time-consuming and error-prone. A For loop provides an efficient way to automate repetitive insert operations. In this program, the loop executes ten times and inserts employee details into the EMPLOYEE table. Employees IDs are generated automatically through the loop counter, employee names are created using string concatenation and salaries are calculated dynamically using an arithmetic expression.

This approach reduces coding effort, improves readability and demonstrates the practical



use of loops in database programming.

Ex:- (i) Variable Declaration

DECLARE

i NUMBER;

A numeric variable i is declared. It is used as loop counter.

(ii) For loop

FOR i in 1..10

The loop executes 10 times, starting from 1 - 10

(iii) Insert statement

INSERT INTO EMPLOYEE

VALUES (i, 'EMPLOYEE-' || i, 2000 + (i \* 1000));

This statement inserts:-

\* Employee ID = Loop counter value

\* Employee Name = EMPLOYEE\_1, EMPLOYEE\_2  
----- EMPLOYEE\_10

\* Salary =  $2000 + (i \times 1000)$

(iv) Output Message

DBMS-OUTPUT.PUT\_LINE ('10 rows inserted successfully');

Displays a confirmation message after successful insertion.

\* Conclusion:- Thus a PL/SQL program using a FOR loop was successfully executed to insert 10 records into the EMPLOYEE table. The For loop simplified the insertion process, reduced repetitive coding and use of iterative statements.



## Practical 8

\* Aim:- Writing SQL and PL/SQL queries to retrieve information from the databases.

\* Theory:- SQL is used to store, manipulate and retrieve data from relational databases. The SELECT statement is one of the most commonly used SQL commands for fetching records from databases tables.

Ex:- In this, table Employee2 is created with attributes Emp-ID, Emp-Name, Salary, Dept.

And the values were inserted using Insert into Employee2 command.

To retrieve information command like

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Select * from Employee2 where salary > 25000;
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(ii) PL/SQL is Oracle's extension of SQL that provides programming features such as variables, loops, cursors and exception handling. A cursor is a PL/SQL construct used to process query results row by row. It helps retrieve and manipulate data from a database in controlled manner.

In this program, PL/SQL cursor fetches each record and displays the employee



name and salary using DBMS\_OUTPUT.PUT\_LINE

Ex:- ① Cursor Declaration

CURSOR emp-cur is

Select Emp-name, Salary from Employee2;  
Creates a cursor that stores the result of the Select query.

② For loop:-

For emp-rec IN emp-cur loop

Fetches one record at a time from the cursor.

③ DBMS\_OUTPUT.PUT\_LINE ('Employee Name:'  
|| emp-rec. Emp-Name || 'Salary:' ||  
emp-rec.Salary);

\* Conclusion :- Thus, SQL and PL/SQL queries were successfully used to retrieve information from the Employee2 table. SQL was used for data retrieval, while PL/SQL cursor processing was used to access and display records one by one.



## Practical 9

- \* Aim:- For a given set of relation schemes, create tables and perform the following Join Queries Inner Join, Outer Join Subqueries - with IN clause, with Exists clause.
- \* Theory:- In a relational database, data is often distributed across multiple tables to reduce redundancy and maintain data integrity. To retrieve related information stored in different tables, SQL provides various operations such as Joins and Subqueries.
  - A Join is used to combine rows from two or more tables based on a related column. An Inner Join returns only those records that have matching values in both tables, whereas an Outer Join returns matching records along with non-matching records from one or both tables depending on the type of outer join used.
  - A subquery is a query written inside another SQL query. It helps retrieve data based on the results of another query. The IN clause is used to compare a value with a list of values returned by a subquery. The EXISTS clause checks whether the subquery returns any rows and returns True if at least one matching record



exists.

In this practical, Employee1 and Department1 tables are created and different join and subquery operations are performed to analyze the relationships between the data stored in these tables.

(i) Inner Join

Select Employee1.emp-name, Department1.dept-name  
from Employee1 inner join Department1  
on Employee1.deptID = Department1.deptID;

Returns only employees whose department IDs match with a department in the Department1 table.

(ii) Left Outer Join:-

Returns all employees including Sachin, whose department ID is NULL. For unmatched records, the department name is displayed as NULL.

(iii) Right Outer Join:-

Returns all departments including department 106 (MCA), which has no employee assigned. For unmatched departments, employee name is displayed as NULL.

(iv) Subquery with IN clause

Select emp-name, salary from Employee1 where  
deptID IN (Select deptID from Department1  
where dept-name = 'MCA');

The subquery finds all department IDs having the name MCA, and the outer query displays



employees belonging to those departments.

⑤ Subquery with Exists clause:-

Select emp-name from Employee1 E where  
exists (Select \* from Department1 D where  
E.deptID = D.deptID);

Checks whether a matching department exists for each employee and returns only those employees having a valid department.

\* Conclusion:- Thus, various SQL operations such as inner join, Left Outer Join, Right Outer Join, and Subqueries using IN and EXISTS clauses were successfully performed on the Department1 and Employee1 tables to retrieve related information from the database efficiently.